

Computer-based gait analysis of dogs: Evaluation of kinetic and kinematic parameters after cemented and cementless total hip replacement

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Keywords

Total hip replacement, gait analysis, treadmill, kinetic analysis, kinematic analysis

Summary

To date it is unclear whether cementless total hip replacement (THR) in dogs is of clinical advantage in comparison to cemented THR with regard to lameness improvement. Thus the aim of this study was to compare objectively the development of the gait pattern after cemented and cementless THR in dogs. For this purpose, 18 adult dogs with hip dysplasia underwent computer-based gait analysis on an instrumented treadmill prior to unilateral THR and then again ten days, four weeks and four months after surgery. Analysed kinetic parameters were symmetry indices (SI) of vertical ground reaction forces (GRF), which included peak vertical forces (PFz), mean vertical forces

(MFz), vertical impulse (IFz), and vertical ground reaction forces of the arthroplasty limbs only. Analysed kinematic parameters were range-of-motion and the flexion and extension angles of hip, stifle and hock joints. The symmetry indices for PVF, MFz and IFz decreased to a value less than six in both THR groups four months after surgery, which is defined as not lame. Improvement in lameness of the arthroplasty limbs during the examination period of four months was not significantly different between the cemented and cementless groups. The results suggest that within a short-term observation period of four months after surgery, neither cementless nor cemented THR have a greater advantage with regard to lameness improvement. Additional studies with larger pools of subjects and longer time periods for follow-up examinations are necessary to verify these findings.

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Introduction

Hip dysplasia is a frequently diagnosed orthopaedic problem in dogs, even though many breed organizations try to minimize the prevalence of this disease by breed selection. A common treatment of hip dysplasia is total hip replacement (THR). The

first unipolar arthroplasty in dogs and cats was reported in 1953 (1). Since the 1970s, THR has become more important for the treatment of painful diseases of the canine hip. In the beginning, only cemented prostheses were implanted. The implantation of cementless prostheses was first carried out in experimental dogs to study the fixation

process and the effects on bone remodelling around the implant (2–4). The first reports of the results of cementless THR in dogs as clinical patients were published in the mid-1980s (5–7). Today, various cemented and cementless THR systems are available for implantation in dogs.

Many studies have evaluated the clinical and radiographic results of different prostheses systems (6, 8–12). Evaluations of outcome have also included telephone interviews with the owners and force plate analysis (6, 9). One study of cementless porous-coated THR, with a minimum of five years of follow-up, found that muscle mass was equal to the contralateral limb in 18 of 24 subjects with contralateral hip dysplasia, whereas the coxofemoral joint range-of-motion in the arthroplasty limb was normal in 21 of 24 subjects (10). Furthermore, radiographic evaluation found that the femoral and acetabular components remained stable and only minimal alteration of the bone-implant interfaces was detectable.

Several studies investigating the gait pattern and weight bearing characteristics of dogs with hip dysplasia before and after THR have been published (13–19). However, an outcome assessment of the gait pattern and weight bearing characteristics comparing cemented and cementless THR using the same methodology has not been published to our knowledge. One study of 127 dogs implanted with cemented THR found that there was a significant improvement of vertical ground reaction forces at about six months after surgery (13). Another force plate analyses of standing dogs found that cementless THR resulted in normal weight bearing, expressed as percen-

tage of body weight, in comparison to the contralateral non-arthroplasty limb at three months after surgery (15).

Kinematic analyses of clinically normal dogs and dogs with hip dysplasia found that hip dysplasia was associated with alterations in the flexion and extension angles of the coxofemoral, femorotibial and tarsal joints (16). It was also found that dogs with hip dysplasia had a greater degree of coxofemoral joint adduction and adduction-abduction range-of-motion in comparison to clinically normal dogs (18).

Studies comparing cemented and cementless THR in humans found patients preferred cemented THR because the recovery period after cementless THR seems to be more painful. However, no differences between cemented and cementless prosthesis in functional outcome were detected (20–22). Both versions of THR result in a higher walking speed, and less perceived pain and exertion during walking compared to the status before surgery (20–22). Another study found that cementless acetabular implants have better clinical results and a lower loosening rate compared to cemented acetabular implants (23). A force plate analysis of experimental dogs after cemented and cementless THR found that there was an earlier return to normal limb function in the cemented THR group (24). A retrospective study of 46 cemented^a and cementless^a THR patients did not find any significant differences between the groups with regard to functional outcome and short-term radiographic scores (25). Visual gait analysis is a subjective method, although graduation of lameness by means of severity scores can be carried out according to standardized protocols (26). Since the human eye is not able to evaluate motion sequences in intervals of milliseconds, minimal grades of lameness are difficult to identify (27). Besides, results of orthopaedic examinations and visual gait analysis depend on the expert knowledge of the investigator. By contrast, computer-based kinetic and kinematic gait analysis allow for objective conclusions about the gait of orthopaedic patients (28–30). Computer-based gait analysis also has variables that

can be difficult to control such as gait speed, handler variations, and the need for multiple repeated runs to achieve an adequate number of acceptable plate strikes. However, the reported treadmill system with four force plates aids in minimizing some of these variables.

Thus, the purpose of the present study was to analyse objectively the short term process of lameness reduction after cemented and cementless THR by means of kinematic and kinetic gait analyses on an instrumented treadmill.

Methods

For this study, 24 adult dogs of different breeds with hip dysplasia were initially selected for unilateral hip replacement. Inclusion criteria were an age of at least 10 months, absence of any clinical signs of disease besides the hip dysplasia, a maximum body weight of 55 kg, and the ability of the dogs to walk on the treadmill. Dogs were excluded from the data analysis if they suffered a complication that could possibly have a significant effect on limb function, such as a femoral fracture or sciatic neuropathy. In total, 18 dogs completed the study; six dogs were excluded due to severe complications. One group of nine dogs was implanted with a cemented hip prosthesis^a and the other nine dogs with a cementless hip prosthesis^a. The selection process for the type of THR was influenced by surgeon bias and their familiarity with either of the systems. Age, sex, height, and temperament of the dogs did not influence the division process.

On the day of hospitalization all dogs underwent physical, orthopaedic and radiographic examinations and the first gait analysis.

Surgical procedure

The THR surgical procedures were carried out by two experienced surgeons who had participated in special training courses. One surgeon (MF) carried out only cemented THR, the other surgeon (AML) the cementless THR.

The radiographs of each dog taken prior to surgery were used to determine the prosthesis size and to assess the degree of hip dysplasia. Therefore, size calibrated radiographs of the pelvis in ventrodorsal position, of the lateral pelvis, and of the lateral femur were prepared (►Fig. 1). Special acetate templates with all available sizes of femoral and acetabular components were used to choose the optimal size of the implant. Grading of hip dysplasia was carried out according to the Orthopedic Foundation for Animals (31). Parameters examined were subluxation, incongruity of femoral head and pelvis, flattening of the acetabulum, insufficient roofing, deformation of femoral head and degree of osteoarthritis. Degree of hip dysplasia did not influence the group division process.

Premedication was carried out with acepromazin^b (0.05 mg/kg IM) and levome-thadon^c (0.6 mg/kg IV). Anaesthesia was induced with propofol^d (1–5 mg/kg) and maintained with isoflurane. Ceftiofur^e (2 mg/kg SC) and carprofen^f (4 mg/kg SC) were administered prior to induction of anaesthesia. All dogs had epidural anaesthesia with morphine^g (0.1 mg/kg) and bupivacaine^h (0.5 mg/kg).

Unilateral hip replacement was performed using a craniolateral approach. The femoral head and neck were excised with an oscillating saw, and the acetabulum was reamed. In cases where a cemented prosthesis was used, the acetabular component was secured in place with polymethylmethacrylate bone cement containing gentamicinⁱ. The femoral canal was reamed with a series of reamers until a trial prosthesis fitted into the canal. The femoral component was secured in the femur with

^b Vetranquil® 1%: Albrecht GmbH, Aulendorf, Germany

^c L-Polamivet®: Intervet Deutschland GmbH, Unterschleißheim, Germany

^d Narcofol® 10mg/ml: CP-Pharma Handelsgesellschaft GmbH, Burgdorf, Germany

^e Excenel®: Pharmacia GmbH, Karlsruhe, Germany

^f Rimadyl® Injektionslösung: Pfizer GmbH, Berlin, Germany

^g Morphine® 10mg/ml: Hexal AG, Holzkirchen, Germany

^h Bupivacain®-RPR-actavis 0.5%: Actavis Deutschland GmbH, Langenfeld, Germany

ⁱ Refobacin®: R & Biomet Bone Cement R, Biomet Austria GmbH, Austria

^a CFX™ (Cemented fixation), BFX™ (Biologic fixation): BioMedtrix, Inc., Boonton, NJ, USA

polymethylmethacrylate bone cement. The cement was mixed manually directly before injection. The instillation of the cement was carried out manually for the acetabular component, whereas the cement for the femoral component was injected with the aid of a cement gun^j. Special femoral stem positioning tips were not used. A suitable sized femoral head was chosen after testing with a trial head. Finally the hip joint was reduced. In cases where a cementless prosthesis was used, reaming and broaching of the acetabulum and femur were carried out until the prepared cavities reached the appropriate size for the individual dog. The matching femoral component was inserted afterwards. Both systems were implanted using special positioning tools^a under the guidance of the assistants. In case of the appearance of femoral fissures intra-operatively, the bone was stabilised with one to five twist knot cerclage wires (1 – 1.2 mm), depending on the dimension of the fissure. Before closing the wound, a microbiological swab was taken from the region around the joint. All dogs were radiographed post-operatively to evaluate the position of the prosthesis.

Postoperative care

Medications administered after surgery were levomethadon^c (0.2 mg/kg t.d.s. SC) for two days, cefalexin^k (25 mg/kg, b.i.d. p.o.) for 10 days, and carprofen^l (4 mg/kg o.p.d p.o.) for 10 days. Two days after surgery dogs were usually discharged from our hospital to the owners. The owners were instructed to keep the dogs on a short leash to restrict movement and to avoid stairs, slippery floors, interaction with other dogs, and playing for a period of eight weeks.

Four weeks and four months after surgery, the pelvis and femur of the dogs were radiographed again in ventrodorsal and mediolateral views. Changes in the position of the prosthesis as well as presence of bone reactions, loosening of the prosthesis and development of osteoarthritis were recorded (► Fig. 1).

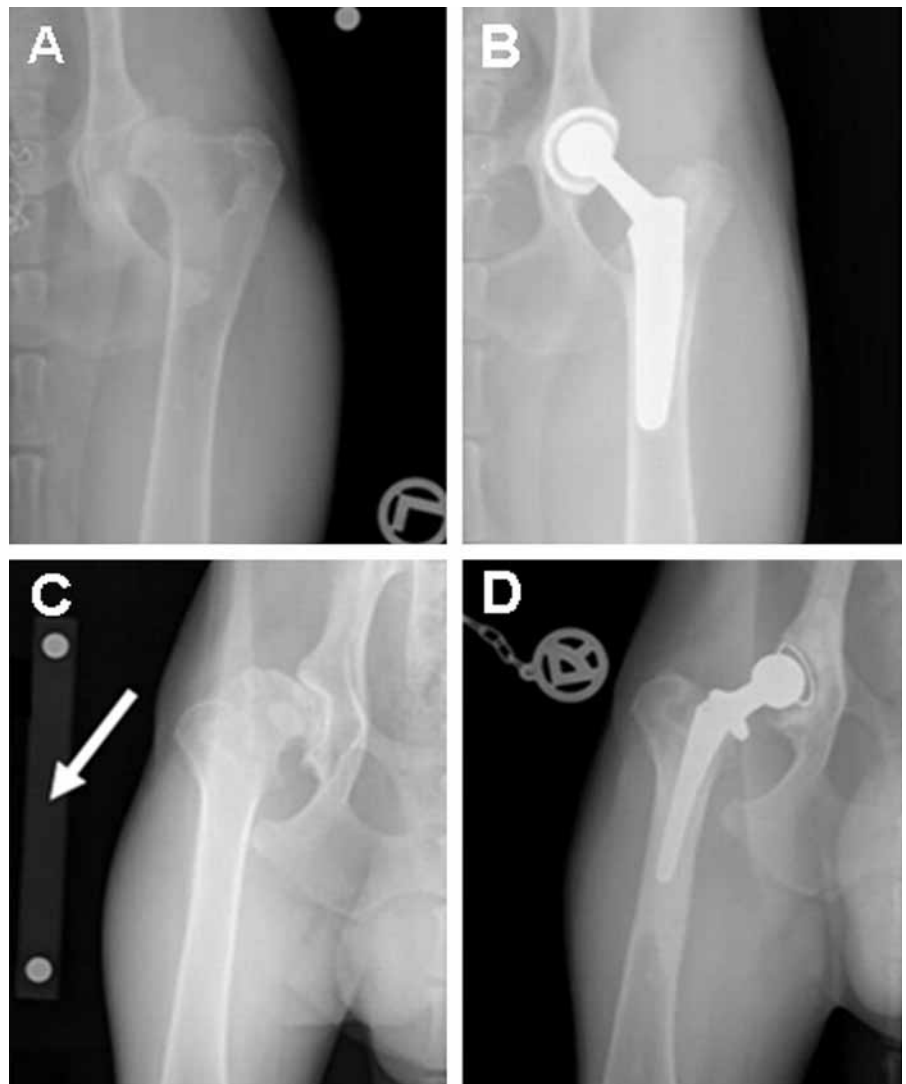


Fig. 1 Example radiographs of a dog before (A) and after (B) cementless total hip replacement, and a dog with cemented total hip replacement respectively (C, D). Both dogs were affected bilaterally by hip dysplasia. **Arrow:** 10 cm calibration tool for the determination of the optimal prosthesis size.

Kinetic and kinematic analysis

One day prior to surgery, the first gait analysis of each dog was performed. Ten days after surgery a second gait analysis was carried out and the sutures were removed. Four weeks and four months after surgery, gait analysis was performed again in all dogs. Analysis of kinetic and kinematic data was carried out using commercial software^m. Measurement frequency was 100 Hz for the kinematic data and 1000 Hz

for the kinetic data, respectively. Before motion capture of the dogs at a gait on the treadmill, static and dynamic calibration of the camera system (maximum measurement error for all four cameras: 0.08 mm) was carried out.

The instrumented treadmillⁿ was a custom-made device with four force plates, each integrated under a separate belt. Accordingly, it was possible to analyse kinetic and kinematic parameters for each hind-limb individually and simultaneously. Each

^j Mathys Ltd, Bettlach, Switzerland

^k Cefalexin® 600: Hexal AG, Holzkirchen, Germany

^l Rimadyl® Kautabletten, Pfizer GmbH, Berlin

^m Vicon Nexus Motion System, Retroreflective Markers: Vicon Motion Systems Ltd, Oxford, UK

ⁿ Model TM-07-B: Bertec Corporation, Columbus, Ohio, USA

belt had a dimension of 30.5 cm width and 1.22 m length. The treadmill was embedded in a wooden platform and covered with carpet; total height was 52 cm.

Motion analysis of the dogs on the treadmill was recorded using four high speed-infrared-cameras^o installed in the corners around the treadmill. For kinematic measurements, retroflective markers^m with a diameter of 16 mm were fixed with double-faced adhesive tape at 16 anatomic landmarks (iliac crest, ischiatic tuberosity, greater trochanter, femur, fibular head, lateral malleolus of the tibia, distal lateral aspect of the fifth metatarsus). Landmarks on the femur and tibia were placed at the mid-diaphysis approximately 2 cm cranial to connecting lines between the greater trochanter and fibular head and between the fibular head and lateral malleolus, respectively. After marker positioning, the dogs were allowed to habituate to the movement on the treadmill. In each session three to six trials of 20 sec were recorded. Walking speed was 0.5 or 0.65 m/s, respectively, depending on the height of the dog and its individual gait behaviour on the treadmill. The recorded retroflective markers appeared as a scatterplot in the commercial software program^m. For the interpretation of kinematic parameters, every marker had to be labelled according to its anatomical localisation. For the calculation of joint angles and other kinematic parameters, a kinematic model of the hindlimb consisting of four segments (pelvis, femur, tibia, and paw) was generated. For the analysis of gait cycle parameters, it was necessary to manually define the time of every foot strike and foot lift-off within a gait sequence. For improved comparability of the movement pattern of the mean joint angle values, the step duration was normalized to a value of 100. Analysed gait cycle parameters in this study were the stance and swing phase of the arthroplasty and contralateral limbs. Flexion and extension of hip, stifle and hock joint angles as well as the range-of-motion of each joint were analysed for each limb separately. All kinematic parameters were filtered with a woltring-filter^m and exported automatically in spreadsheet tables. Ten gait cycles were

analysed for each respective examination time point. In general, kinematic parameters were analysed descriptively.

Regarding the kinetic data, forces in vertical (Fz) and craniocaudal (Fy) direction were analysed. In particular, peak vertical force (maximum load/PFz), mean vertical force (MFz), vertical impulse (IFz), and propulsion and braking force (Fy) were analysed for both hindlimbs separately. For each examination day, 10 valid ground contacts were analysed. A ground contact was defined as being valid when only a hindlimb paw touched the correct belt. That meant that neither of the forelimb paws nor the contralateral hindlimb paw touched the belt and there were no passovers of the analysed hindlimb. Furthermore it was necessary that the dogs walked as normal and straight as possible on the treadmill, with only minimal interference by the handling person. Filtering of the kinetic data was carried out with a Butterworth-filter fourth order with 10 Hz cut-off frequency. All data were converted into percent of body weight (%BW). This translation was necessary for the calculation of a symmetry index between both hindlimbs for the peak vertical force (PVF), mean vertical force (MFz), and vertical impulse (IFz) between the left and right hindlimbs (32). The calculation of the symmetry index was carried out with following formula:

$$\bullet \text{ SIz (\%)} = 100 - [(Fo/Fc) * 100]$$

(SIz = symmetry index of the according parameter (PVF, MFz, IFz); Fo = parameter of the operated hindlimb; Fc = parameter of the contralateral hindlimb)

Symmetry index values of zero percent meant a total symmetry between the two extremities whereas values greater than six percent were defined as lameness, and data less than negative six percent was interpreted as lameness of the contralateral hindlimb. A dog was considered to be free of lameness if all three symmetry indices (SI PVF, SI MFz, SI IFz) were under six percent (33). Symmetry indices calculated at day 10, 30 and 120 after surgery were converted accordingly. In addition to the calculation of the symmetry indices, the vertical and craniocaudal ground reaction forces (in %BW)

of only of the implanted limbs (Fy: implanted and contralateral limb) in both groups were calculated for day 10, 30 and 120 after THR in comparison to those prior to surgery. An analysis of variance (ANOVA) and a Tukey's post test for multiple comparisons were carried out for the PVF, MFz and IFz values of both groups for the calculated symmetry indices and the results of the lameness improvement for only the affected limbs (Fy: implanted and contralateral limb) during the entire examination period.

Values for the statistical comparison of the kinetic data are shown as mean $\pm$ standard error; p-values <0.05 were considered as statistically significant. For the comparison between the cemented and cementless groups, kinetic data of affected limbs were analysed with an unpaired t-test for PVF, MFz and IFz at the time prior to surgery and four months after surgery. To analyse if the cemented and the cementless group differed with regard to body weight and age, data of both groups were analysed with an unpaired t-test.

Results

In total, eighteen dogs of different breeds were enrolled in this study (► Table 1); six dogs were excluded from the study due to severe complications. The cemented group consisted of five male and four female dogs, and the cementless group of seven female and two male dogs. The differences between the cemented and the cementless group for body weight and age were not significant (► Table 1).

Complication rate

In the cemented group, two dogs of the initial 11 dogs were excluded due to severe complications. One of them had a sciatic neuropraxia, the other a luxation of the prosthesis with subsequent sciatic neuropraxia. In the cementless group, four of the initial 13 dogs were not analysed because three of them were implanted with a hybrid prosthesis due to femoral stem loosening (one dog), acetabular component loosening (one dog), and hair-line fracture

^o High-speed, infrared cameras, MX 3+: Vicon Motion Systems Ltd, Oxford, UK

of the femur (one dog). In one dog, the prosthesis had to be removed after a femoral fracture.

Complications in the analysed dogs of the cemented group were a femoral fissure in one dog, and a luxation seven weeks after THR surgery in another dog. Overall complication rate in the analysed dogs of the cemented group was 22%.

Complications among the analysed dogs in the cementless group were a femoral fissure in one dog, a luxation of the femoral head five days after surgery in another dog, and a mild, short-term sciatic neuropraxia in two dogs. Overall complication rate in the analysed dogs of the cementless group was 44%.

Radiographs

Radiographic images obtained prior to surgery showed that all but two dogs (one dog in each group with a contralateral hip dysplasia score of 4) were affected by bilateral hip dysplasia (► Table 1). Both of the unilaterally affected dogs had major clinical signs of pain due to severe hip dysplasia, for which treatment by THR was recommended. The preoperative hip dysplasia scores for the arthroplasty limb prior to surgery were 5 ($n = 1$), 6 ($n = 6$), and 7 ($n = 11$) (► Table 1).

Four weeks and four months after surgery, no dog had radiographic signs of loosening or bone reaction. All prostheses had remained in their original position and there were not any changes in the development of osteoarthritis in the contralateral hindlimbs (► Fig. 1). Subjective assessment of the radiographs indicated signs of minimal subsidence of the femoral stem in two dogs of the cementless group. Periosteal reaction, bone lysis or changes in the interface quality could not be detected in the periprosthetic bone in any of the 18 dogs.

Kinetic data

Vertical ground reaction forces

Comparison of vertical ground reaction forces (in %BW) of the implanted limbs between the cemented and cementless groups was carried out for the time points prior to

Table 1 Breed, weight, gender, age, type of prosthesis system used, and hip dysplasia (HD) score (31) of the dogs included in the study. M = male, F = female, MN = male neutered, FS = female spayed, 5 = mild hip dysplasia, 6 = moderate hip dysplasia, 7 = severe hip dysplasia.

Breed	Body weight (kg)	Gender	Age (years)	Hip dysplasia score
Cemented				
Mixed breed	35	MN	10	7
Mixed breed	27	FS	9	6
Mixed breed	27.2	MN	7	7
Mixed breed	46	FS	3	7
Mixed breed	38.5	MN	3	6
German Shepherd	31.5	F	2	7
German Shepherd	28	F	2	6
Labrador	35	MN	2	5
Šarplaninac	55	M	11 months	7
Mean $\pm$ SE	35.9 $\pm$ 9.4		4.3 $\pm$ 3.1	
Cementless				
Mixed breed	37.5	M	3	7
Mixed breed	30.5	FS	4	7
Labrador	24	F	10	7
Labrador	26.5	FS	4	6
Labrador	29.5	F	4	6
Bernese Mountain dog	42.5	FS	6	6
Bernese Mountain dog	43	F	1	7
Boxer	27	FS	3	7
Kangal	33.7	F	11 months	7
Mean $\pm$ SE	32.7 $\pm$ 7		4 $\pm$ 2.8	

Key: SE = standard error; M = male; F = female; MN = male neutered; FS = female spayed; for hip dysplasia score. 5 = mild hip dysplasia; 6 = moderate hip dysplasia; 7 = severe hip dysplasia.

surgery and four months after surgery. The differences between the cemented and cementless groups at both time points for PVF, MFz and IFz were not significant (► Fig. 2).

Considering the changes in vertical ground reaction forces within each group during the four month period (► Fig. 2), there were significant improvements for PVF. In both groups, the PVF significantly ($p < 0.05$) improved in the 10 days following surgery (cemented group: mean 30.34 ± 1.84 %BW; cementless group: mean 28.51 ± 2.44 %BW) and four months after surgery (cemented group: 37.04 ± 0.71 %BW;

cementless group: 37.56 ± 1.34 %BW). In the cementless group, the MFz values differed significantly between the preoperative (19.07 ± 1.71 %BW) and four months postoperative values (24.83 ± 0.84 %BW; $p < 0.05$), and between 10 days (17.77 ± 1.79 %BW) and four months postoperatively ($p < 0.01$). For the IFz, significant improvements were found in the cementless group four months postoperatively (19.69 ± 0.88 %BW) in comparison to preoperative (13.3 ± 1.29 %BW; $p < 0.01$) and 10 days postoperative values (12.34 ± 1.41 %BW; $p < 0.001$).

Symmetry indices

Baseline symmetry index values prior to surgery of the two groups were different. The symmetry indices of the cementless group started at a higher lameness level than the cemented group, but this difference was not statistically significant. Curve

progression in symmetry indices suggests worsening in lameness ten days after surgery followed by a progressive improvement in lameness. In general, data showed a high variance. In both groups and for all three symmetry indices, symmetry index values at four months postoperatively for PVF, MFz and IFz decreased to less than six,

which is defined as 'not lame' (► Fig. 3). However, the only significant changes in symmetry index with time were in the cementless group between the preoperative and four months postoperative values for PVF (23.25 ± 6.25 versus 3.19 ± 2.51 [$p < 0.05$]), MFz (20.45 ± 5.57 versus 1.0 ± 2.35 [$p < 0.1$]) and IFz (25.1 ± 6.27 versus -2.99 ± 3.49 [$p < 0.01$]). In the cemented group, none of the changes in symmetry index between the preoperative and 4 month postoperative values for PVF (11.58 ± 5.86 versus 3.35 ± 2.83), MFz (10.15 ± 5.84 versus 1.62 ± 3.16) and IFz (12.17 ± 6.76 versus 1.26 ± 4.71) were significant. The differences between the cemented and cementless groups for all three indices four months after surgery were not significant.

Craniocaudal ground reaction forces

There were hardly any differences between the groups or within each group in the y-forces. Braking force of the cemented THR hindlimb was $4.31 \%BW$ prior to surgery and $3.46 \%BW$ four months after surgery. In the cementless group, the braking force was $3.96 \%BW$ prior to surgery and $3.51 \%BW$ four months after surgery. Small changes were seen for the propulsion force in the arthroplasty hindlimbs. Prior to surgery, these propulsion forces were $2.68 \%BW$ in the cemented group and $3.79 \%BW$ in the cementless group. After four months, the propulsion forces were $5.68 \%BW$ in the cemented group and $5.95 \%BW$ in the cementless group. The y-force data of the contralateral hindlimb in both groups showed no significant differences throughout the observation period within the limb and in comparison to the arthroplasty limb.

Kinematic data

Evaluated gait cycle parameters were the stance phase and the swing phase. None of these parameters showed relevant changes, neither inside one group during the evaluation period nor between groups prior to or after surgery.

Joint angles were analysed descriptively for the flexion and extension of hip joint, stifle joint and hock, as well as for the range-of-motion (ROM) of the respective joints. For all joint angles, slight changes

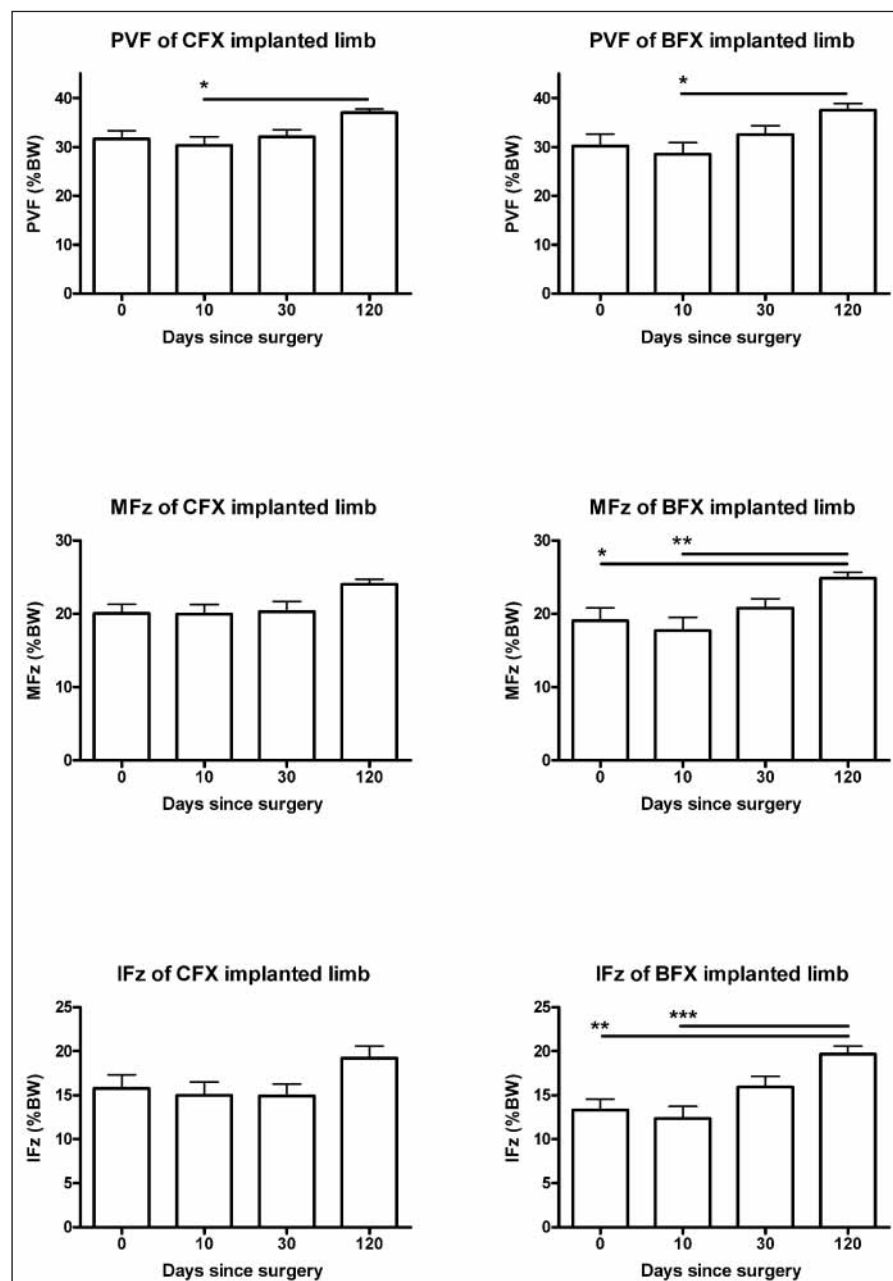


Fig. 2 Mean $\pm$ standard error of peak vertical force (PVF), mean vertical force (MFz), and vertical impulse (IFz) in % body weight (BW) prior to surgery, and at 10 days, four weeks, and four months after cemented (CFX) and cementless (BFX) total hip replacement. Statistical analysis was carried out with an analysis of variance and a Tukey's post test for multiple comparisons (* = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$).

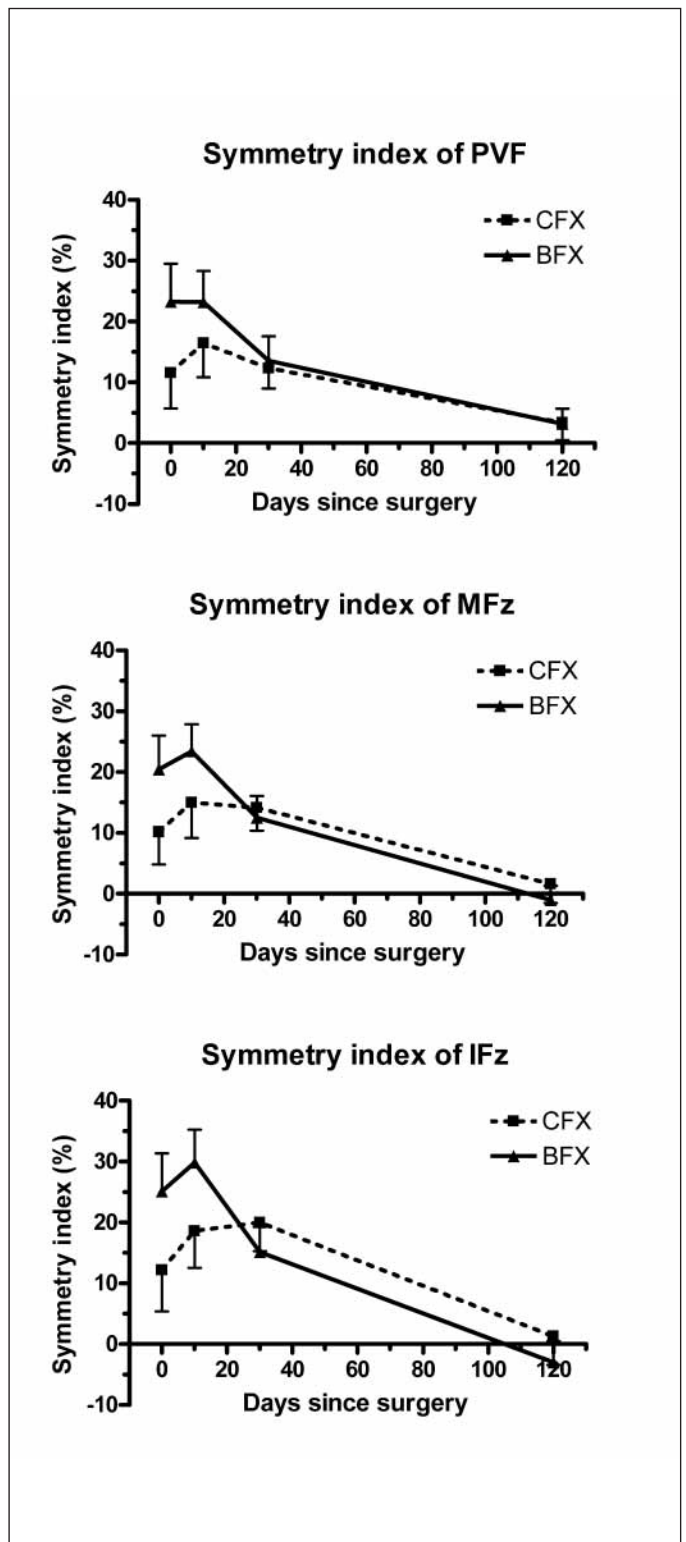
were seen (►Fig. 4). However, these changes in joint angles were not analysed statistically. Hip joint angles prior to surgery ranged from 119.57° to 106.03° in the cemented group (ROM = 13.54°), and from 127.78° to 111.04° (ROM = 16.74°) in the cementless group. Four months after surgery, values ranged from 110.7° to 130.15° (cemented: ROM = 19.45°), 106.81° to 127.41° (cementless: ROM = 20.6°), respectively. Data of stifle joint angles prior to surgery were 102.6° to 128.13° (ROM = 25.53°) for the cemented group and 105.18° to 130.9° (ROM = 25.72°) for the cementless group. Four months after surgery, angles of 94.44° to 125.01° (ROM = 30.57°) for the cemented group and 104.31° to 132.16° (ROM = 27.85°) for the cementless group were analysed. In the cemented group, hock angles ranged from 99.33° to 118.18° (ROM = 18.85°) prior to surgery and from 95.52° to 118.7° (ROM = 23.18°) four months after surgery. The cementless group shows values of 104.65° to 121.96° (ROM = 17.31°) prior to surgery, and 101.65° to 126.07° (ROM = 24.42°) four months after surgery for the hock.

Discussion

The aim of the present study was to evaluate differences in the limb function recovery process, especially in the gait pattern, within a four month period after cemented and cementless THR in the dog. The calculation of symmetry indices for the ground reaction forces in axial direction allowed for the comparison of the loading conditions within a pair of extremities (32). One disadvantage of this index is that all dogs of the present study, except for two, had bilateral hip dysplasia, thus the index compared the arthroplasty limb with an abnormal contralateral limb. Furthermore, worsening or improving disease progression in the contralateral limbs might have affected symmetry index values in our study. Nevertheless, the symmetry index is currently the only suitable method to show an overall decrease in lameness. To rule out a false interpretation of the lameness progression by means of the symmetry index, the progression of only the arthroplasty limb was also analysed. The data in the

Fig. 3

Comparison of symmetry indices of peak vertical force (PVF), mean vertical force (MFz), and vertical impulse (IFz) (mean $\pm$ standard error) during the evaluation period of four months after cemented (CFX) and cementless (BFX) total hip replacement.



present study shows that both groups had symmetry indices above six percent in all three parameters (PVF, MFz, IFz) prior to surgery, meaning that all dogs were lame.

Four months after surgery, the symmetry index values were less than six percent in both groups. However this improvement in the symmetry index was only significant

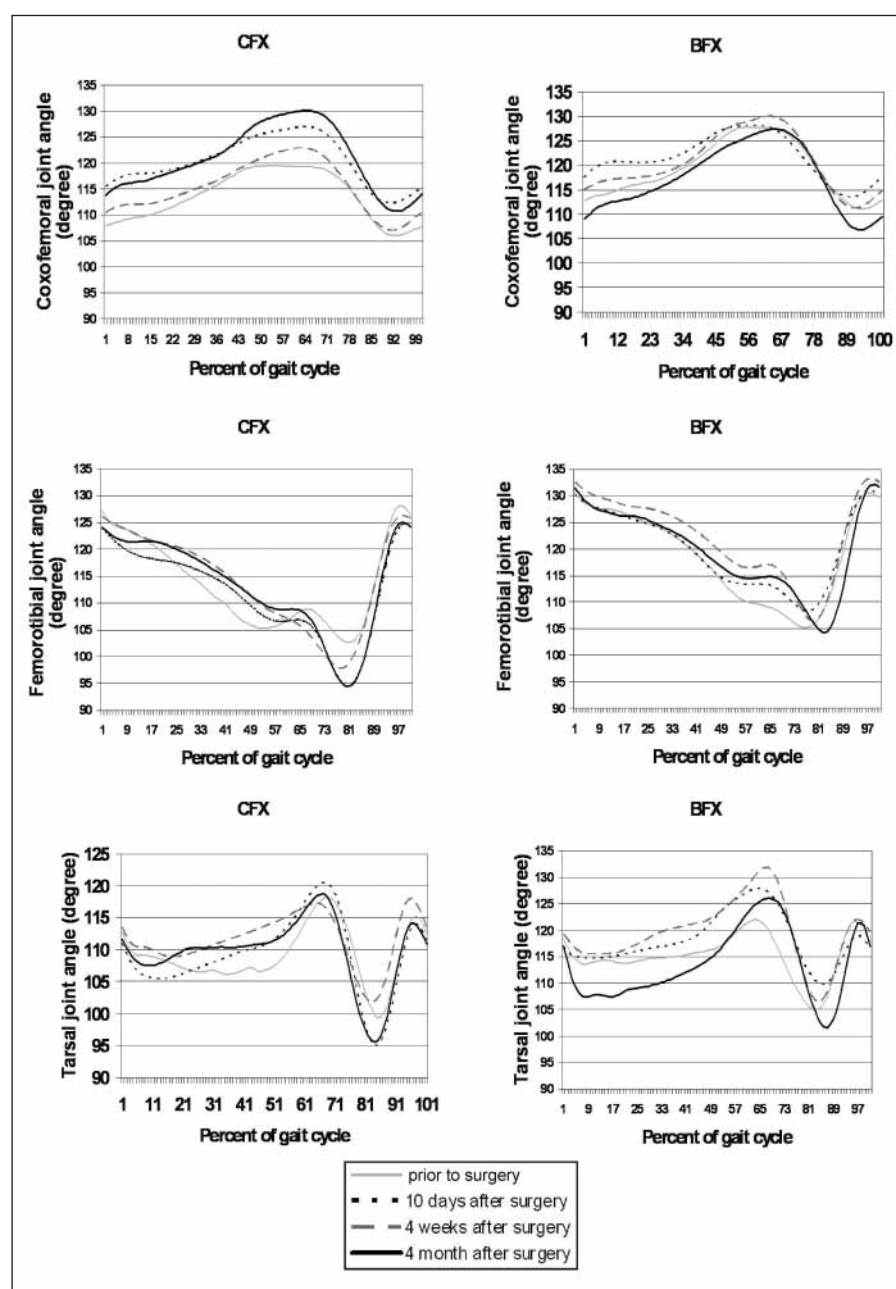


Fig. 4 Description of mean time-averaged curves kinematic evaluation of joint angles during a four month evaluation period after cemented (CFX) and cementless (BFX) total hip replacement.

for the cementless group between prior to surgery and at four months after surgery. This might be due to the high variance in the kinetic data of both groups and the fact that the cementless group started from a higher lameness level prior to surgery than the cemented group. For that reason it is difficult to draw a direct comparison of the lameness improvement between the groups. However, both groups showed a

near to normal limb use four months after surgery. In contrast to our findings, one previous study found an earlier return to normal limb function in experimental dogs after cemented THR in comparison to dogs implanted with a cementless hip implant (24). Due to the fact that healthy research dogs were used for this study, and the cementless prosthesis system was different to the one used by us, the findings of this ex-

perimental study are not comparable to the findings of our study.

Our joint angle analyses are in agreement with values obtained by Kosfeld who performed a retrospective study of the clinical and radiographic outcome as well as gait analysis of dogs after THR (34). Furthermore the stifle joint angle data obtained by us are in agreement with the results of another study that analysed the gait pattern of the Alsatian dog with the help of a treadmill (30). Nevertheless, we found that there were only slight changes during the examination period and also only small differences between the two groups. One limitation of our study is that the kinematic data were only analysed descriptively. Another restriction of our kinematic analyses is that a heterogeneous pool of subjects is not an optimal starting point for kinematic examinations because of the different anatomic distinctions of the different breeds. Dogs affected by hip dysplasia mainly have motion impairments and signs of pain while standing up or laying down, jumping, walking on slippery ground, or after playing intensively. A severe lameness can occur with any degree of hip dysplasia, but does not always correlate with severe radiographic osteoarthritis scores (35). Although there was a significant improvement in some kinetic parameters preoperatively to four months postoperatively in both groups, only slight improvement could be found in the kinematic parameters. From these findings we conclude that it is not necessarily the fact that kinematic parameters during gait are affected in dogs with severe hip dysplasia. More obvious changes in kinematic parameters might be present during other movements of the hindlimb. Therefore, in further studies it would be interesting to analyse kinematic parameters of the hindlimb prior to and after THR during a sitting down or standing up movement.

A concerning aspect of our study is the high complication rate in the cemented and the cementless groups. The outcome of both cemented and cementless THR surgery is subject of many evaluations (6, 8, 9, 36, 37). The complication rates in studies that examined a larger number of patients (84–258 prostheses) were considerably lower than in our study. However, studies

with fewer subjects had higher complication rates. For example, three studies of 28, 37, and 23 THR had complication rates of 50%, 32.43% and 39.13%, respectively (38–40). Due to the normal learning curve of surgeons who work with new systems, complication rates decrease by increasing experience, which could be one reason for the high complication rate in our study. Nevertheless, analysis of the complication rate was not subject of the present study. Only dogs with mild complications like bone fissure or temporary neuropraxia were investigated further by gait analysis. However, even these mild complications could have affected lameness progression in the concerned dogs of both groups, especially at the time of first gait analysis after surgery. Other limiting factors of this study are the small number of subjects and the short evaluation period. Further studies with a larger pool of dogs and later follow-up examinations should be carried out to verify the established data. Furthermore it should be kept in mind that the findings of this study are not universally valid. Other cemented or cementless THR procedures as well as the individual experience of the respective surgeon can result in different lameness outcomes.

In this study, a heterogeneous pool of subjects with several different breeds was used. This pool represents the average of the patients admitted to our clinic. The main interest of our study was the investigation of the general outcome with regard to lameness progression for the two types of THR prostheses and not the comparison of the outcome in different breeds. Furthermore, in case of the kinetic parameters, a heterogeneous pool is very common (24, 25, 41).

The differences of the group populations, such as the assignment of German Shepherds to the cemented group, might have influenced the groups disparity. The selection process used in this study was influenced by the findings of a recent study which suggested that a 'stovepipe' femoral morphology, as described for German Shepherds, is a risk factor for fracture after cementless THR (42). In this context, it has to be kept in mind that not all surgeons follow these selection criteria. Another explanation for differences between the groups

could have been changes in the bone implant interface, periosteal reactions, or bone lysis. However, presence of periprosthetic radiographic changes which could have supported this hypothesis were not evident four months after surgery. As a limitation of the present study, it has to be mentioned that the quality of the radiographic examination might not have been sufficient enough to detect small implant position shifts and small changes in bone morphology.

Our findings of a lack of significant differences in craniocaudal ground reaction forces, either between groups or within the groups, were in agreement with those of Manley et al. who also did not find significant differences between cemented and cementless hip prostheses for this parameter (24). In healthy dogs, the magnitude of forces in the craniocaudal direction is in the range of two to 12% of body weight, which is similar to the values obtained by us for dogs with hip dysplasia and following THR (33, 43, 44).

In the present study, an advantage of either the cementless or the cemented THR system with regard to a short-term improvement in lameness reduction could not be identified. Kinetic analysis showed an almost similar improvement of vertical GRF four months after surgery in the cemented and cementless THR groups. In both groups, there were no evident changes or differences for kinematic parameters found. Decision-making for selection of patients for cemented or cementless THR depends on more than kinetic or kinematic values. Age, temperament, bone-structure, anatomic shape, secondary disorders, owner compliance, and last but not least experience of the surgeon also play a very prominent role in this context.

In conclusion, the choice between a cemented or a cementless hip prosthesis, must be made based on the individual case and should take into account all of the above mentioned parameters.

Conflict of interest

None declared.

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